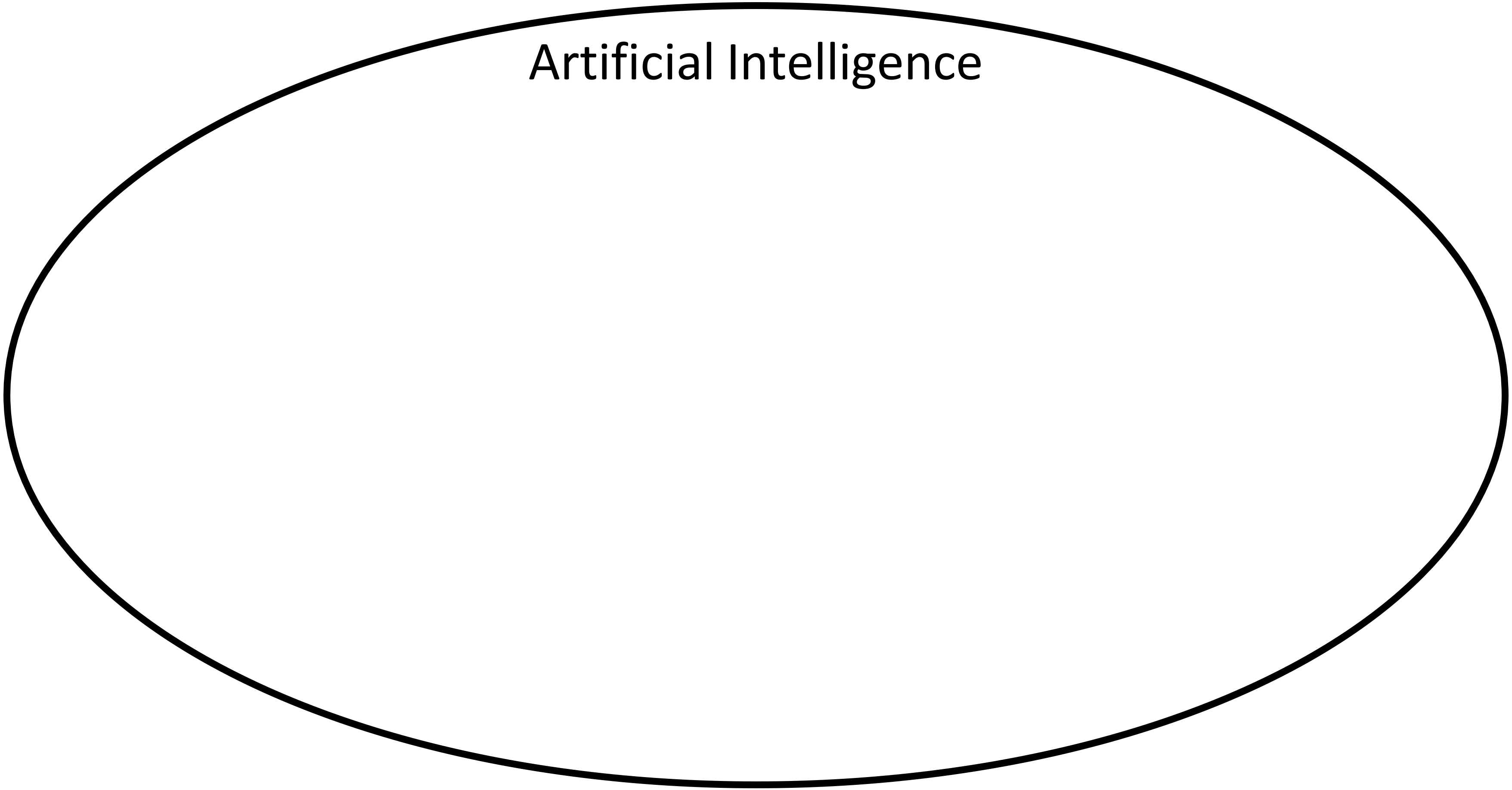
Introduction to Computer Vision

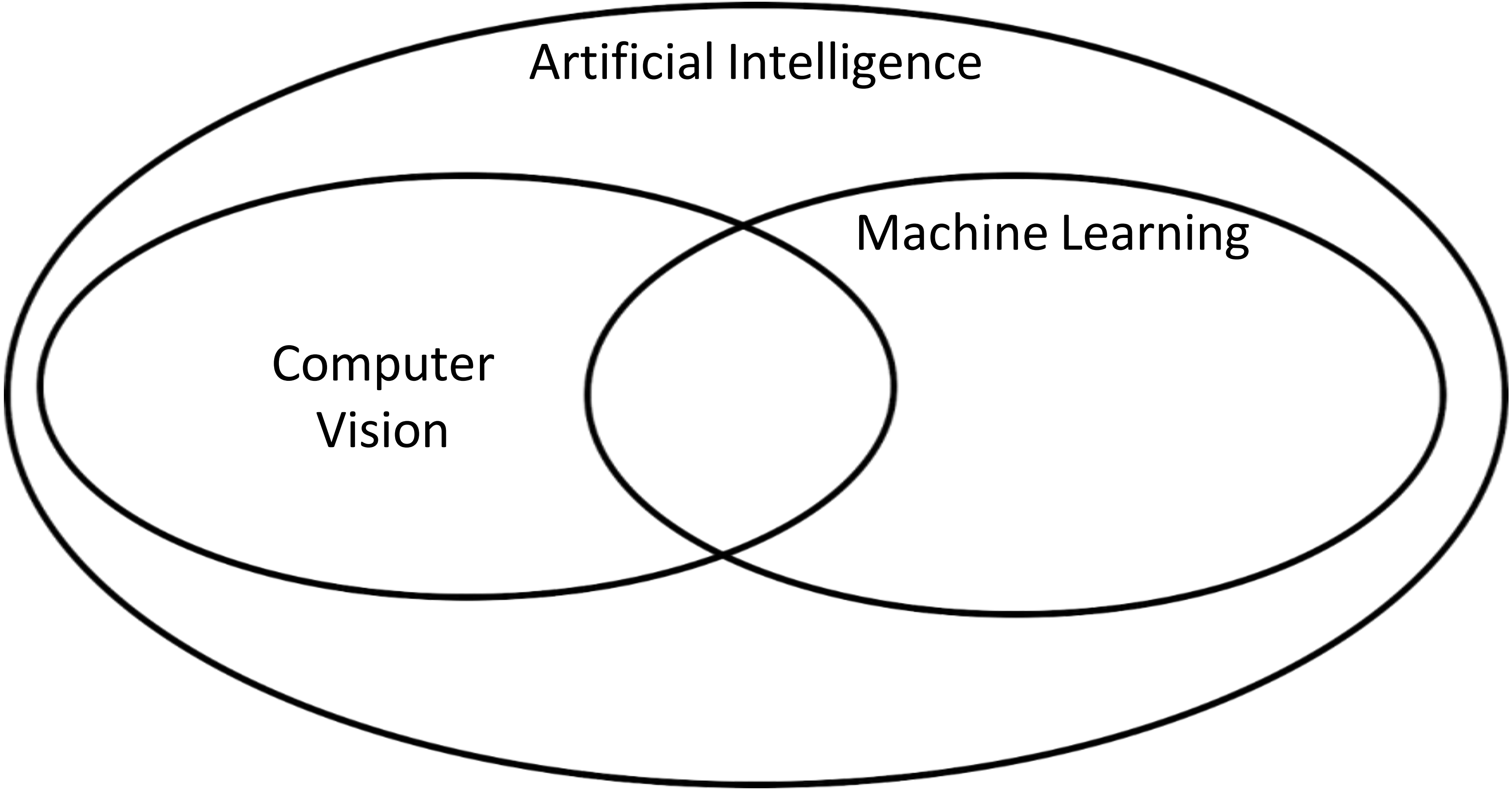
Course Code: CSC 4254 Course Title: Computer Vision & Pattern Recognition

Dept. of Computer Science
Faculty of Science and Technology

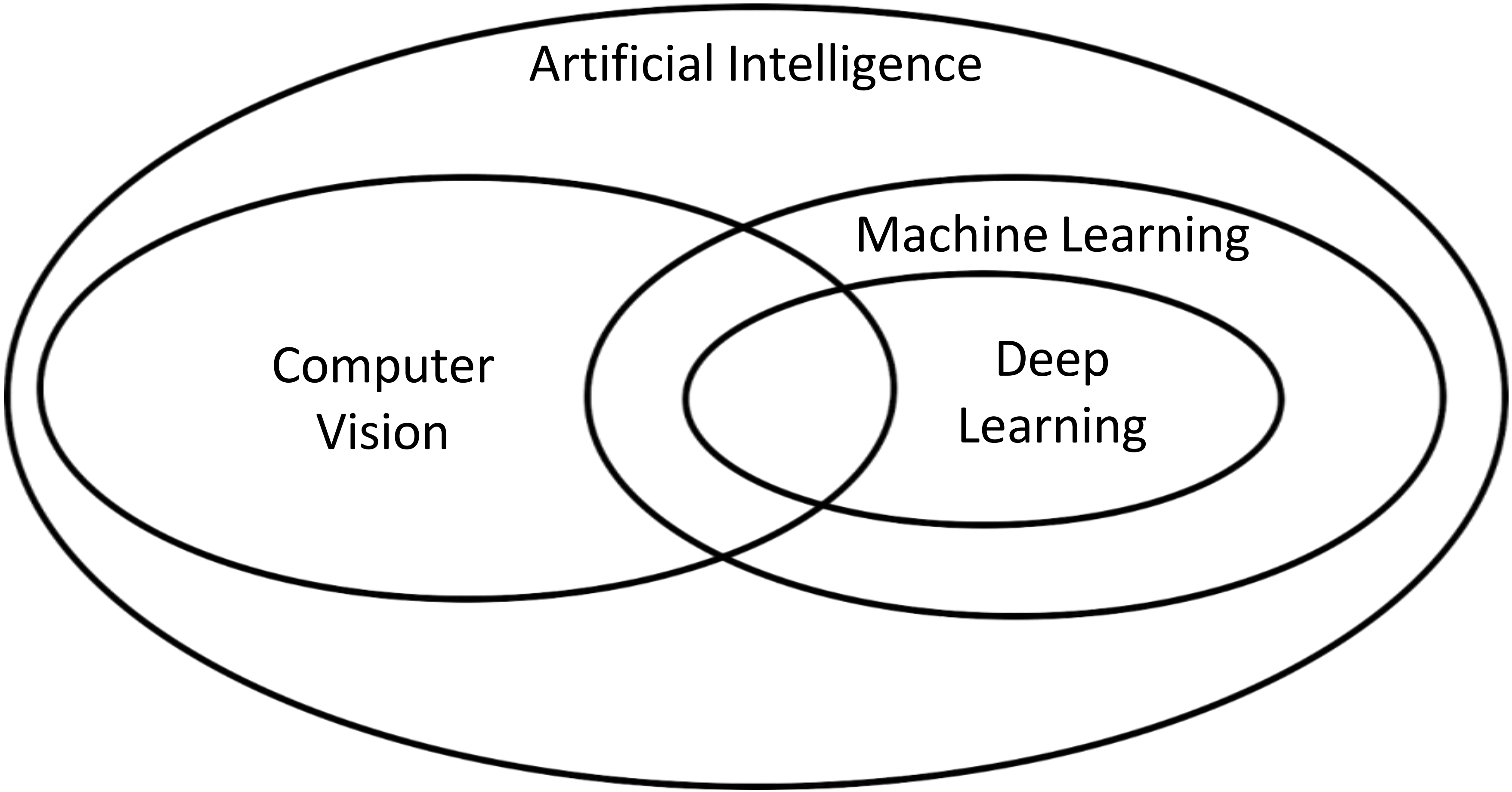
Lecture No:	**1**	**Week No:**	**1**	**Semester:**	
Lecturer:					



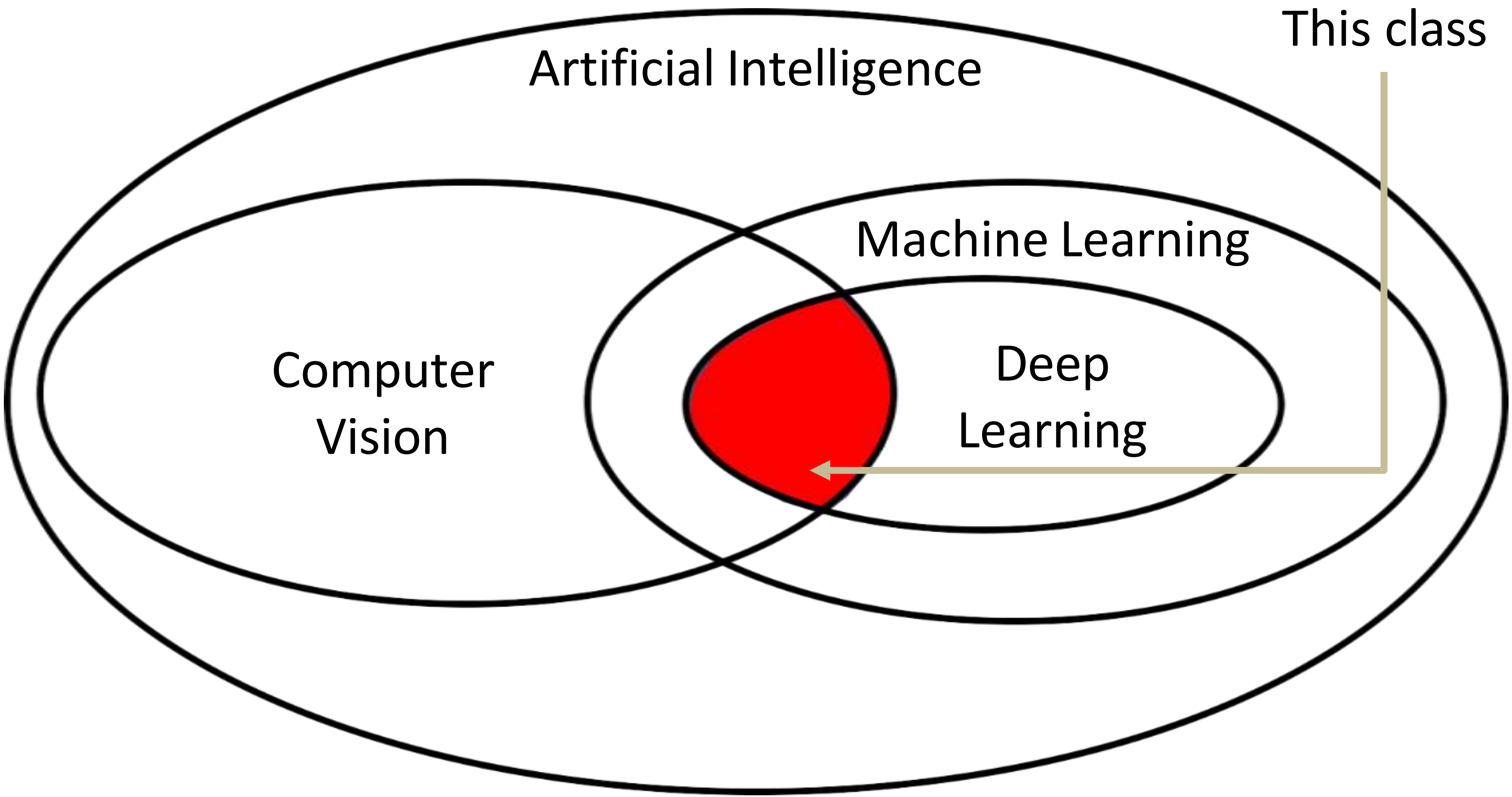
Slide inspiration: Justin Johnson



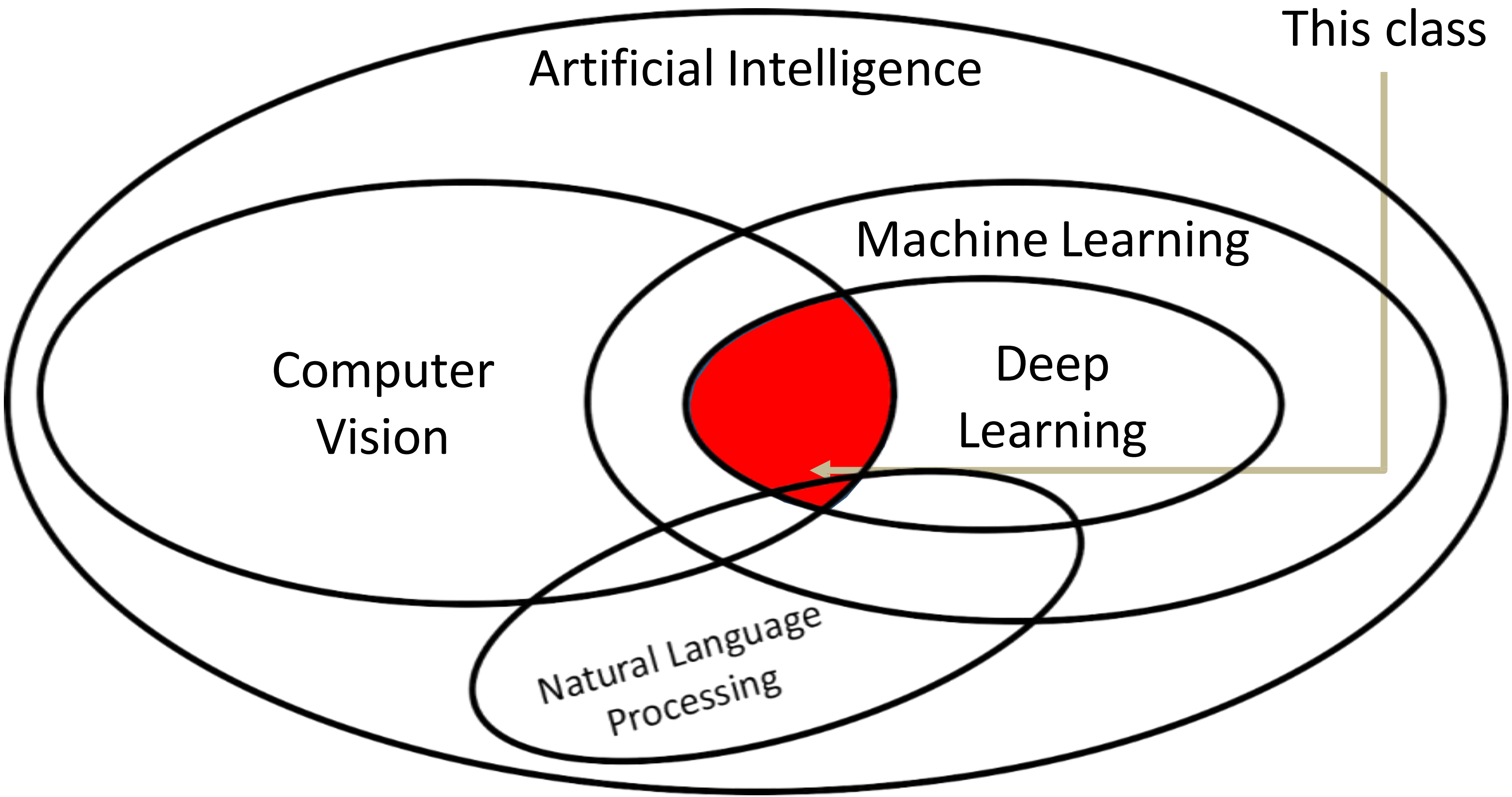
Slide inspiration: Justin Johnson



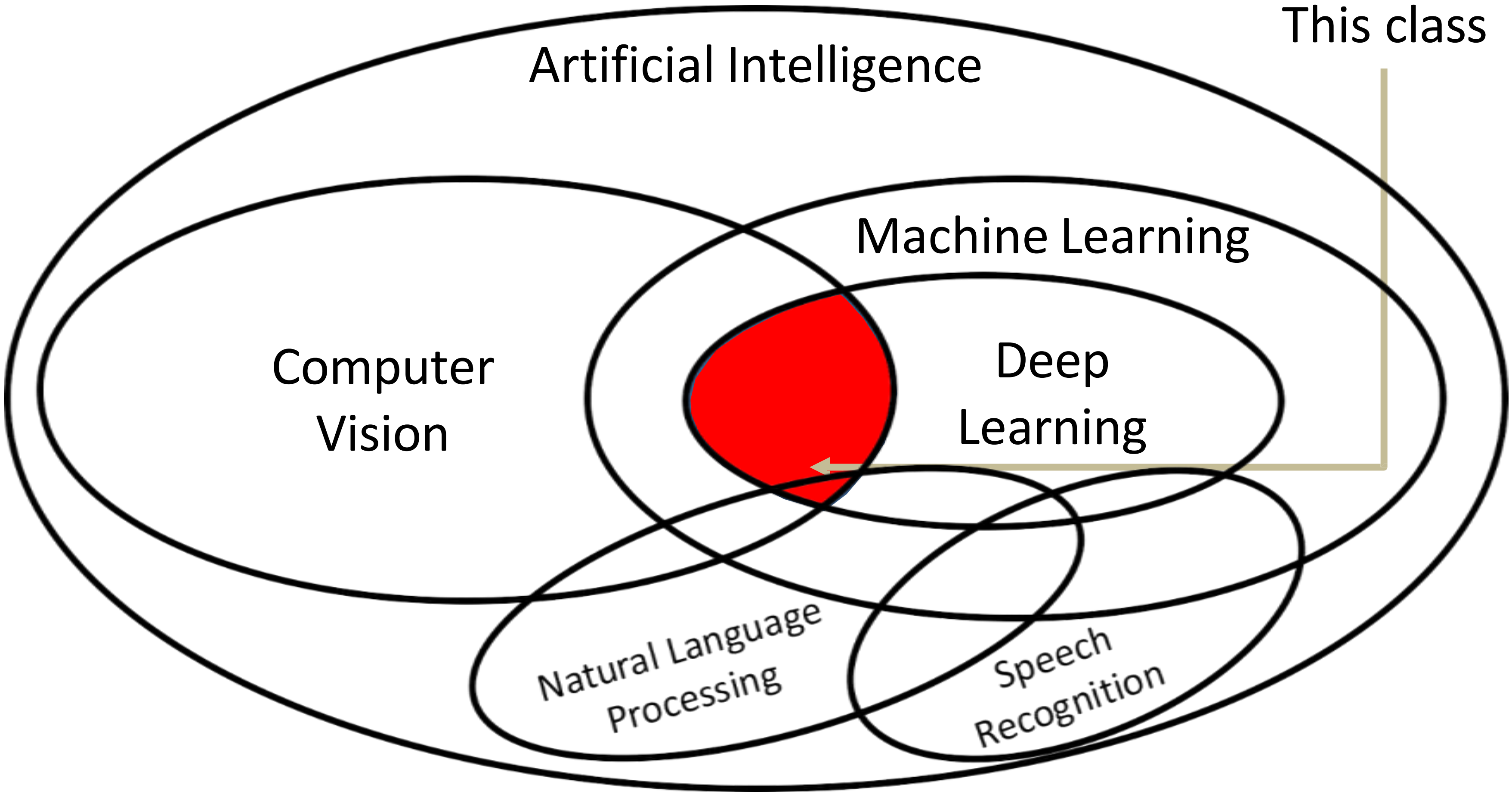
Slide inspiration: Justin Johnson



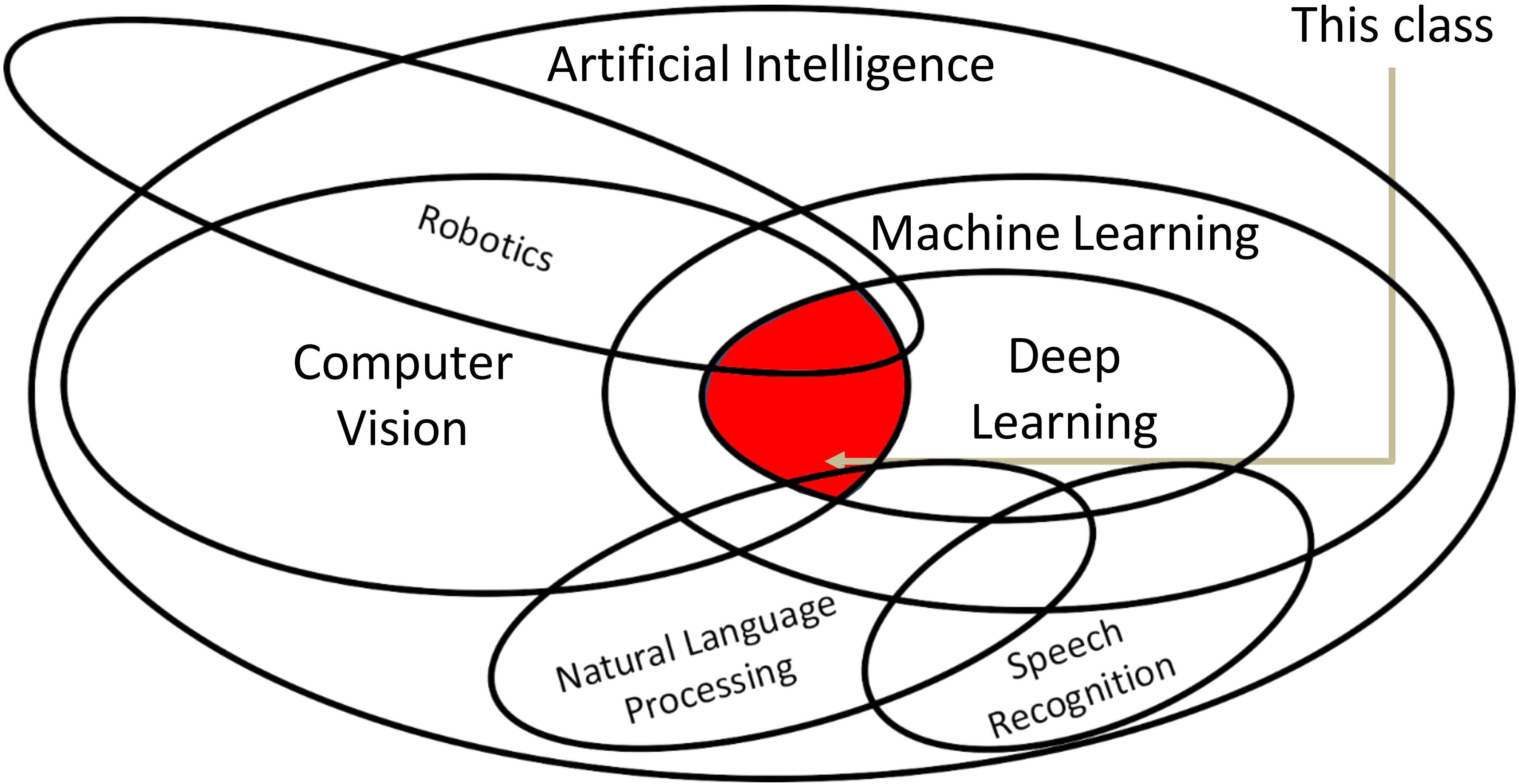
Slide inspiration: Justin Johnson



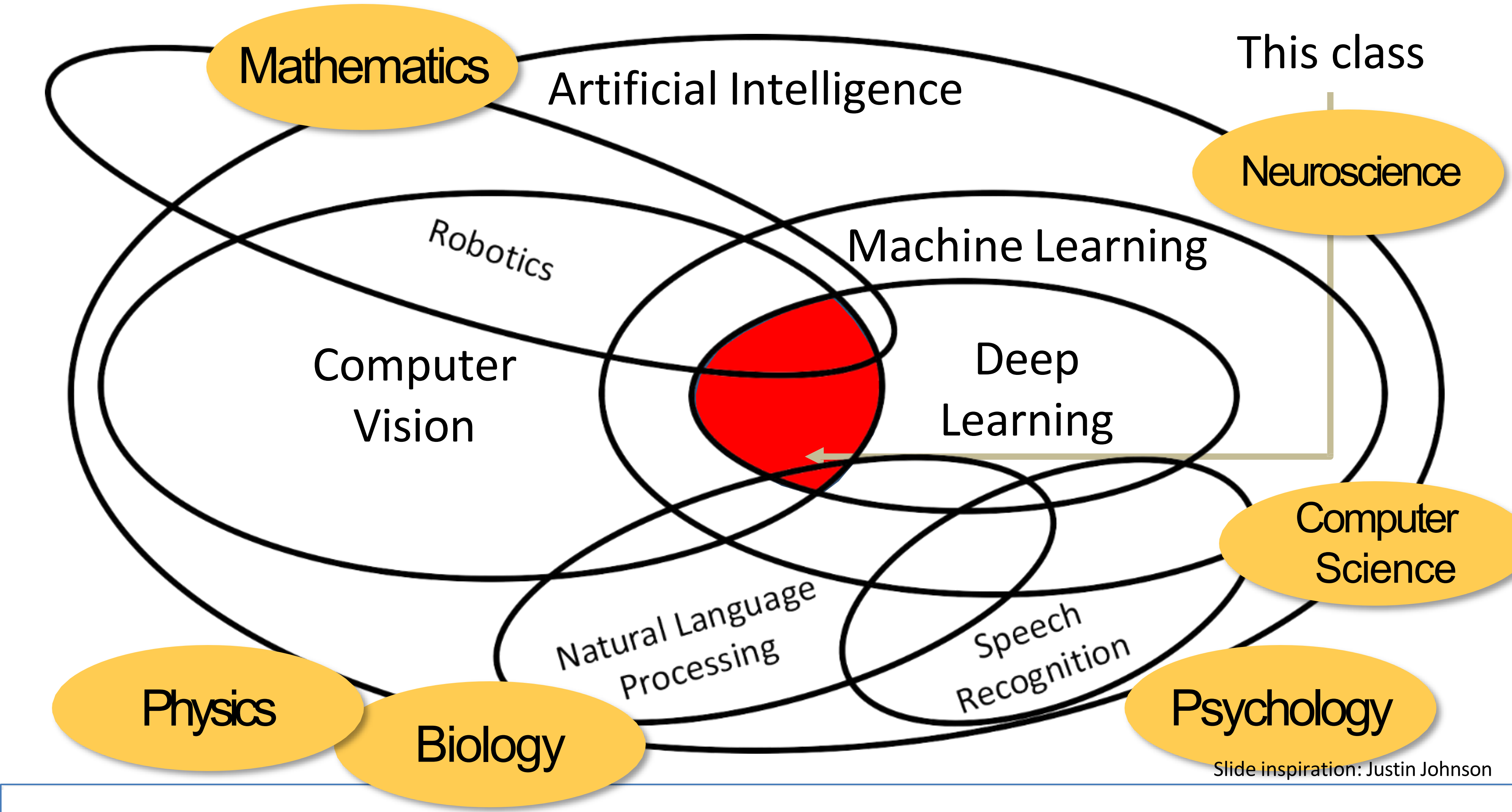
Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson



Mathematics
Artificial Intelligence
This class
Neuroscience
Robotics
Machine Learning
Computer Vision
Deep Learning
Computer Science
Natural Language Processing
Speech Recognition
Physics
Biology
Psychology
Slide inspiration: Justin Johnson

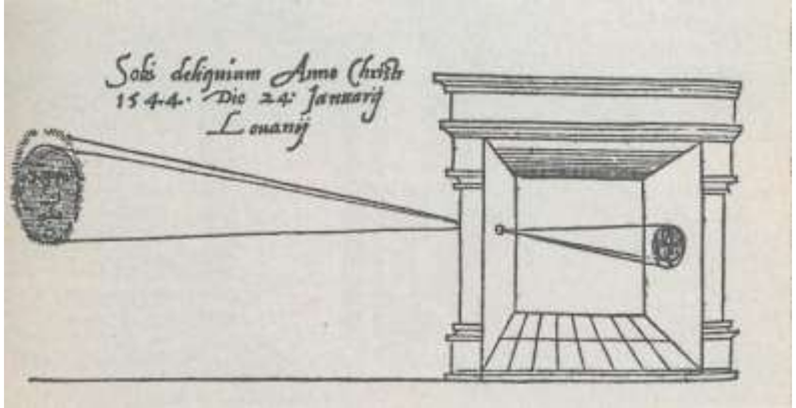
Today's agenda

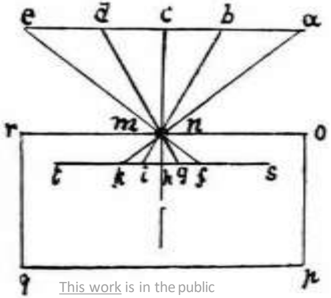
- A brief history of computer vision and deep learning
- Course overview

Evolution's Big Bang:
Cambrian Explosion, 530-540million years, B.C.

Camera Obscura

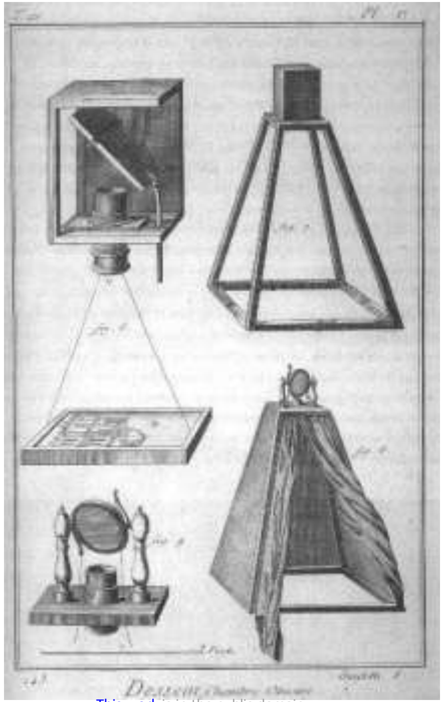
Gemma Frisius, 1545





Leonardo da Vinci, 16th Century AD

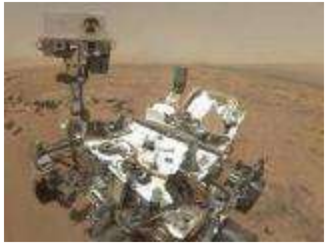
Encyclopedia, 18th Century



Computer Vision is everywhere!





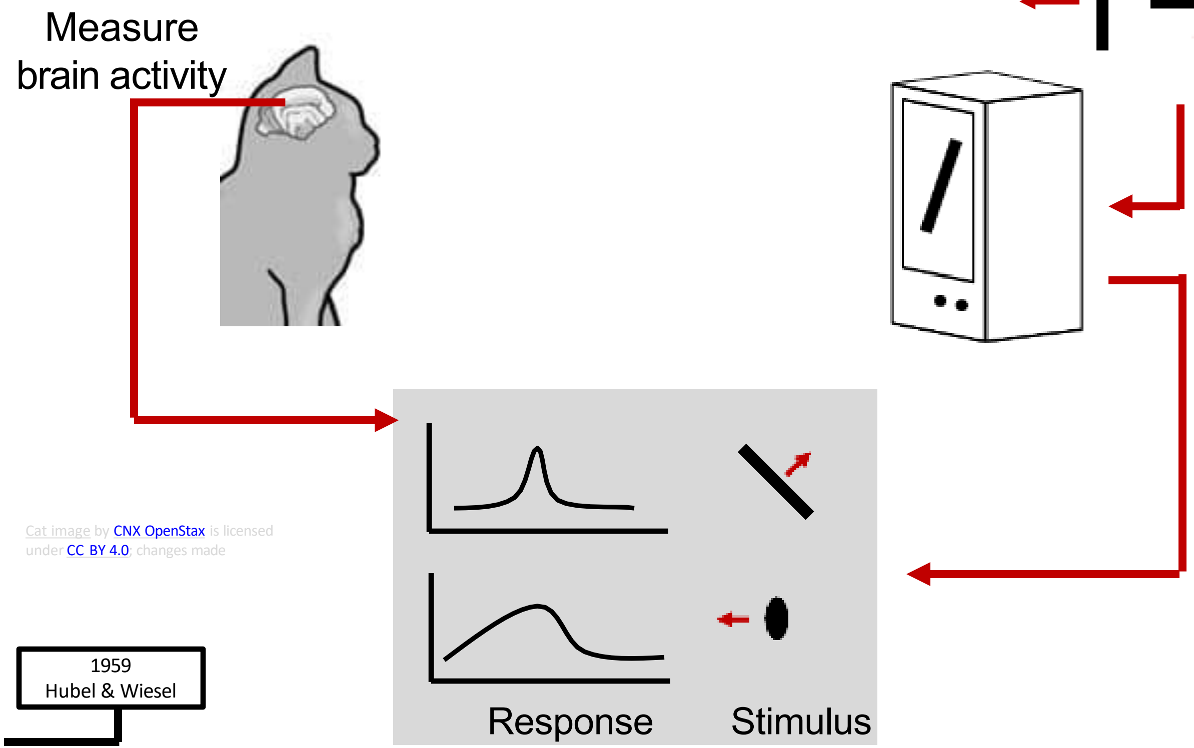



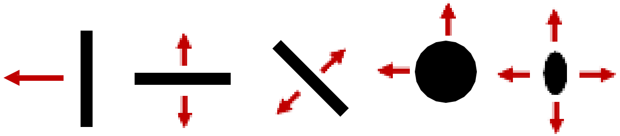




Where did we come from?

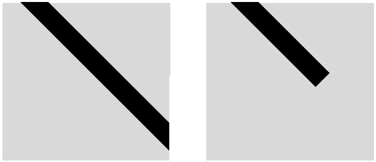
Hubel and Wiesel, 1959





Simple cells:
Response to specific
rotation and orientation

Complex cells:
Response to light
orientation and
movement, some
translation invariance

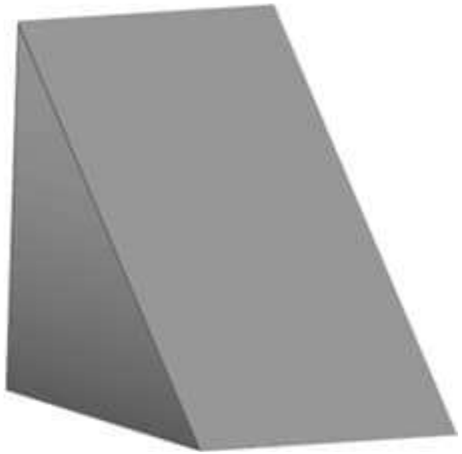


No response

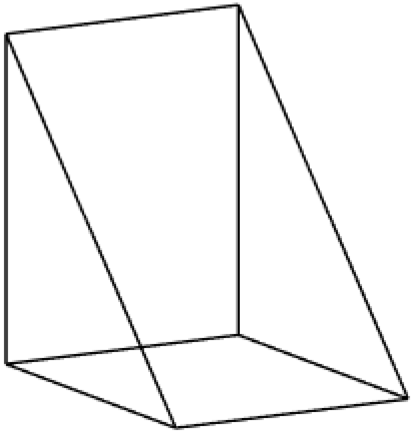
1959
Hubel & Wiesel

Slide inspiration: Justin Johnson

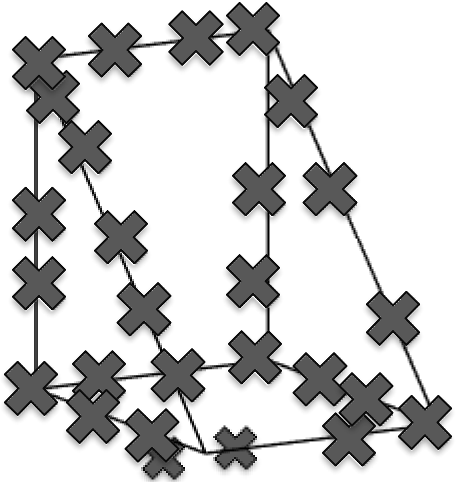
Larry Roberts, 1963



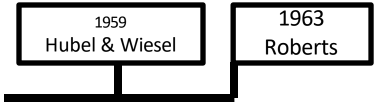
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



Lawrence Gilman Roberts, “Machine Perception of Three-Dimensional Solids”, 1963

Slide inspiration: Justin Johnson

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

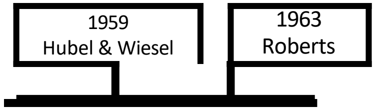
Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

Seymour Papert

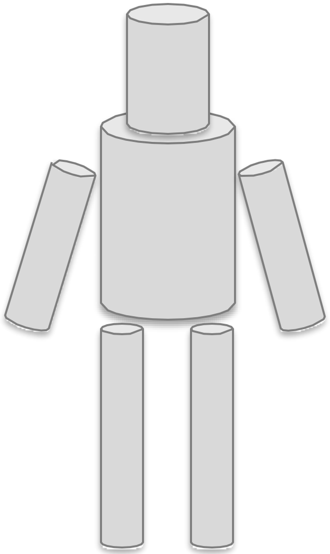
The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".



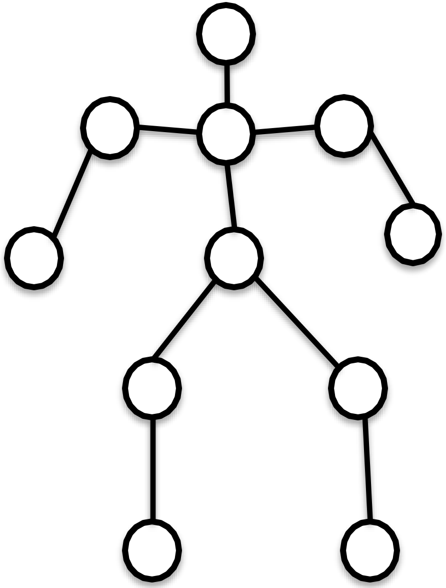
https://dspace.mit.edu/handle/1721.1/6125

Slide inspiration: Justin Johnson

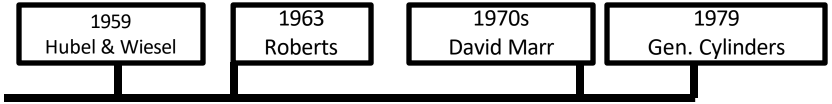
Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford,
1979



Pictorial Structures,
Fischler and Elshlager, 1973



Slide inspiration: Justin Johnson

Recognition via Edge Detection (1980s)

1959 Hubel & Wiesel | 1963 Roberts | 1970s David Marr | 1979 Gen. Cylinders | 1986 Canny

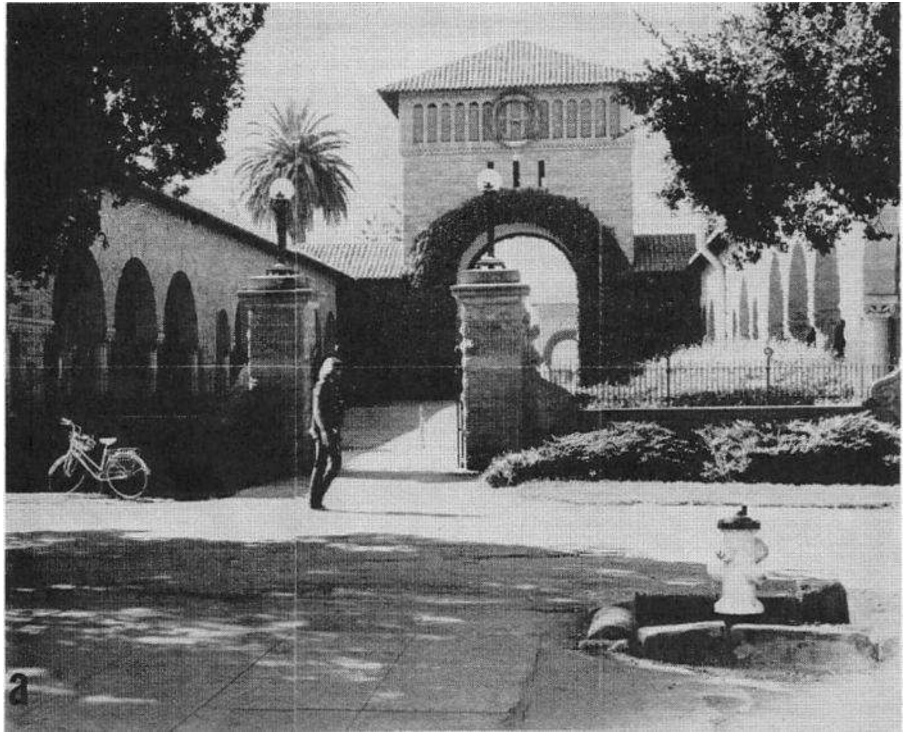
John Canny, 1986
David Lowe, 1987

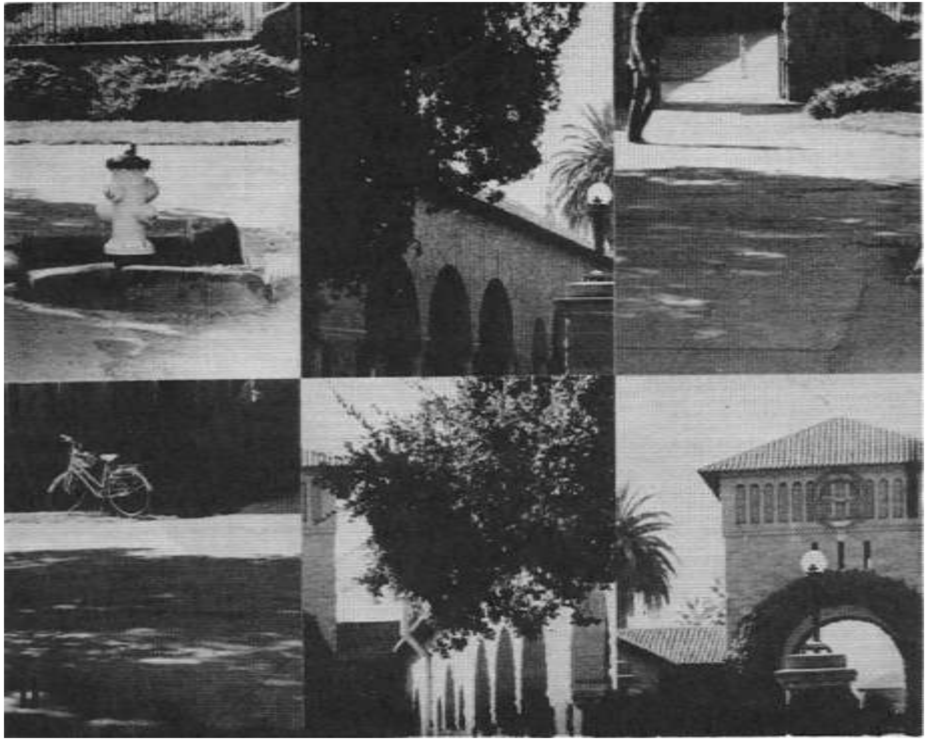
Slide inspiration: Justin Johnson

In the meantime…seminal work in cognitive and neuroscience

Perceiving Real-World Scenes

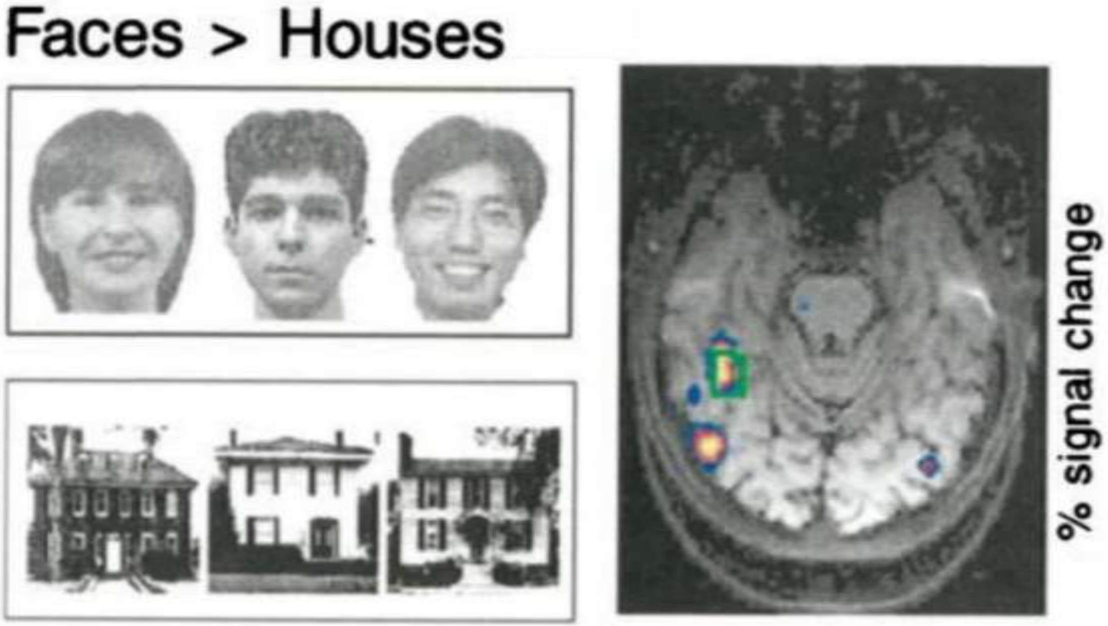
Irving Biederman



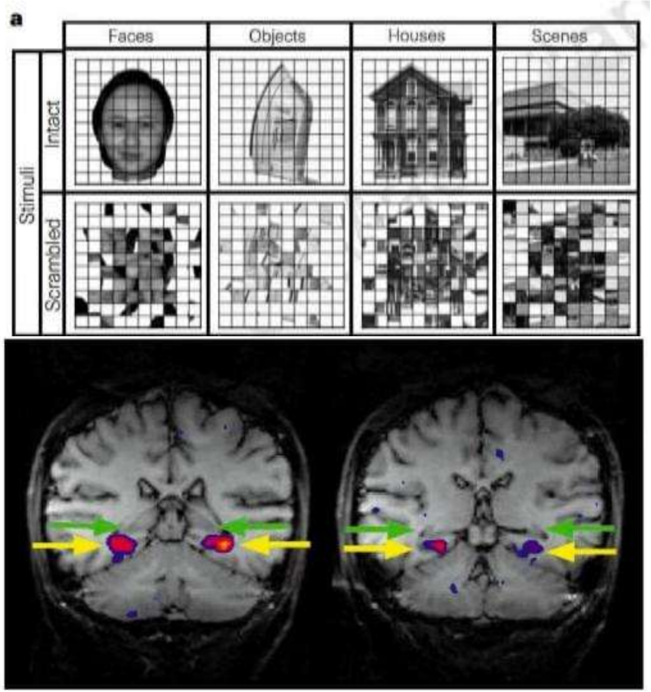


I. Biederman, *Science*, 1972

Neural correlates of object & scene recognition

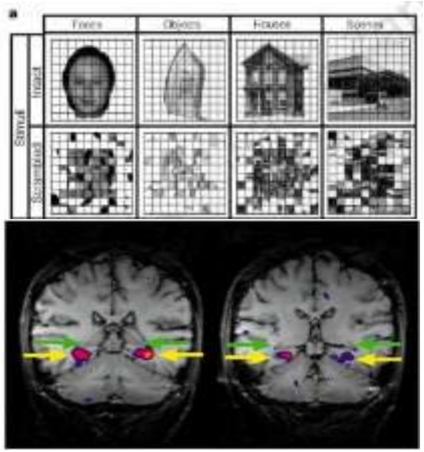


Kanwisher et al. J. Neuro. 1997



Epstein & Kanwisher, Nature, 1998

Visual recognition is a fundamental task for visual intelligence



Recognition via Grouping (1990s)

1959
Hubel & Wiesel

1963
Roberts

1970s
David Marr

1979
Gen. Cylinders

1986
Canny

1997
Norm. Cuts

AI Winter

Normalized Cuts, Shi and Malik, 1997

Slide inspiration: Justin Johnson

Recognition via Matching (2000s)

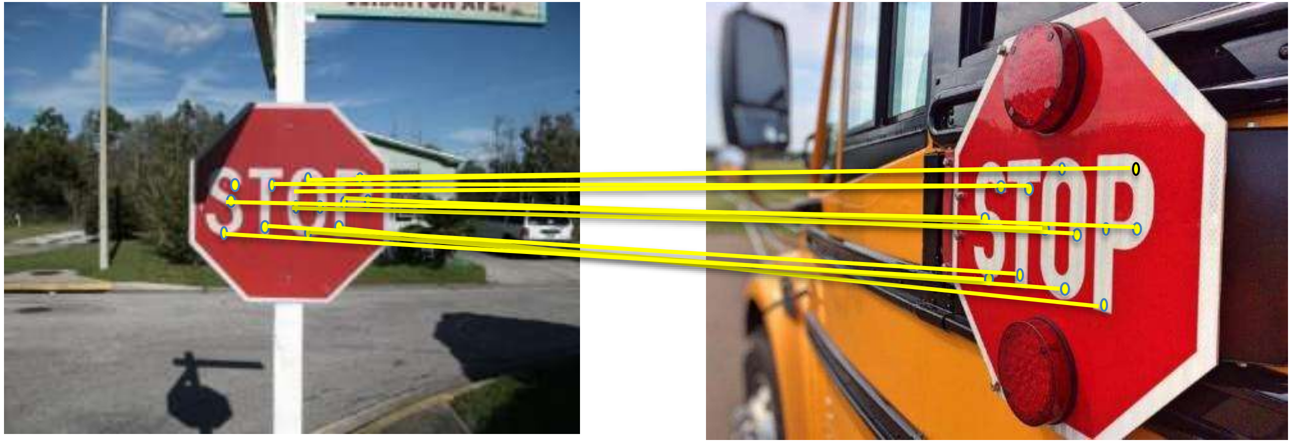


Image is public domain

Image is public domain

1959 Hubel & Wiesel

1963 Roberts

1970s David Marr

1979 Gen. Cylinders

1986 Canny

AI Winter

1997 Norm. Cuts

1999 SIFT

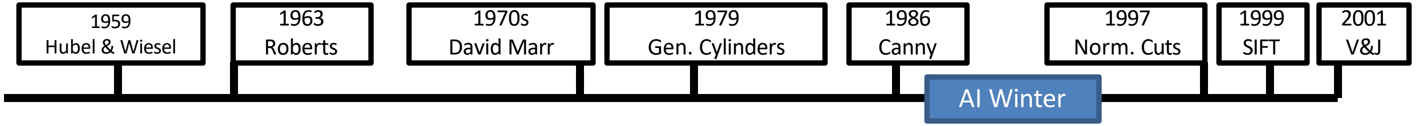
SIFT, David Lowe, 1999

Slide inspiration: Justin Johnson

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision



Slide inspiration: Justin Johnson

Caltech 101 images

Pictures of objects belonging to 101 categories. About 40 to 800 images per category. Most categories have about 50 images. Collected in September 2003 by Fei-Fei Li, Marco Andreetto, and Marc'Aurelio Ranzato. The size of each image is roughly 300 x 200 pixels.

PASCAL Visual Object Challenge

Image is CC0 1.0 public domain

Image is CC0 1.0 public domain

1959 Hubel & Wiesel — 1963 Roberts — 1970s David Marr — 1979 Gen. Cylinders — 1986 Canny — AI Winter — 1997 Norm. Cuts — 1999 SIFT — 2001 V&J — 2004, 2007 Caltech101; PASCAL

Slide inspiration: Justin Johnson

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images

Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

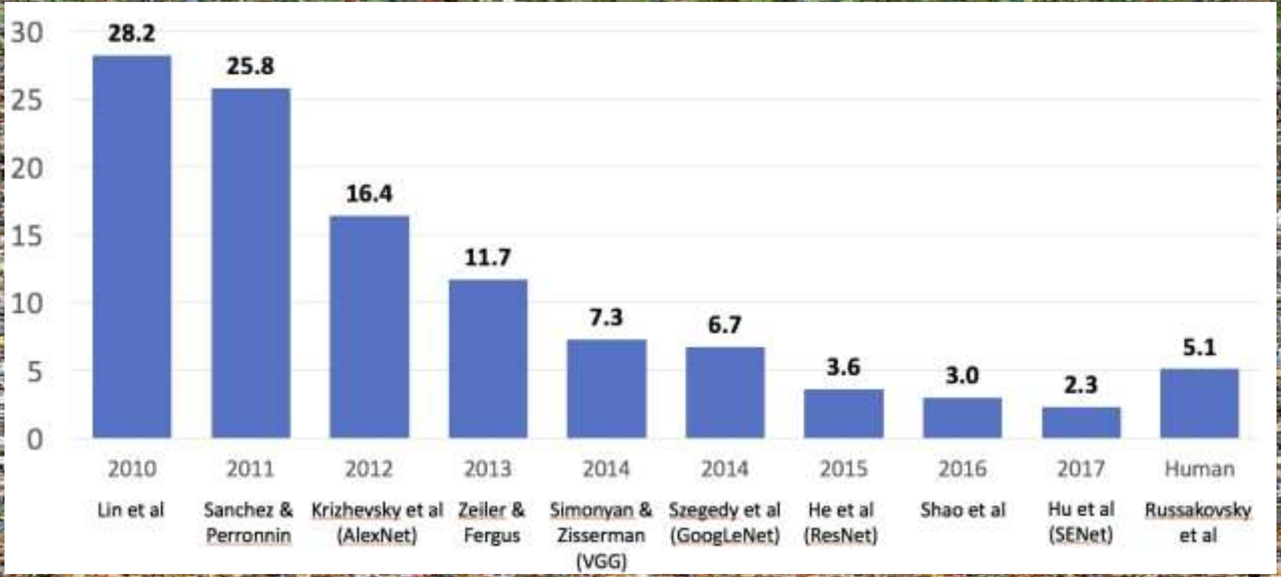
Deng et al, 2009
Russakovsky et al. IJCV 2015

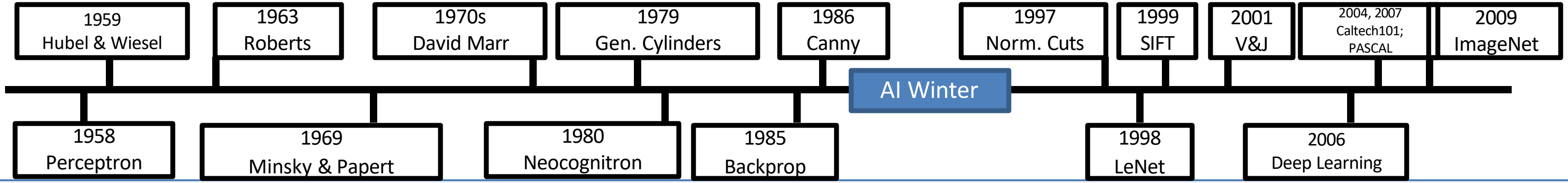
1959 Hubel & Wiesel
1963 Roberts
1970s David Marr
1979 Gen. Cylinders
1986 Canny
1997 Norm. Cuts
1999 SIFT
2001 V&J
2004, 2007 Caltech101; PASCAL
2009 ImageNet

AI Winter

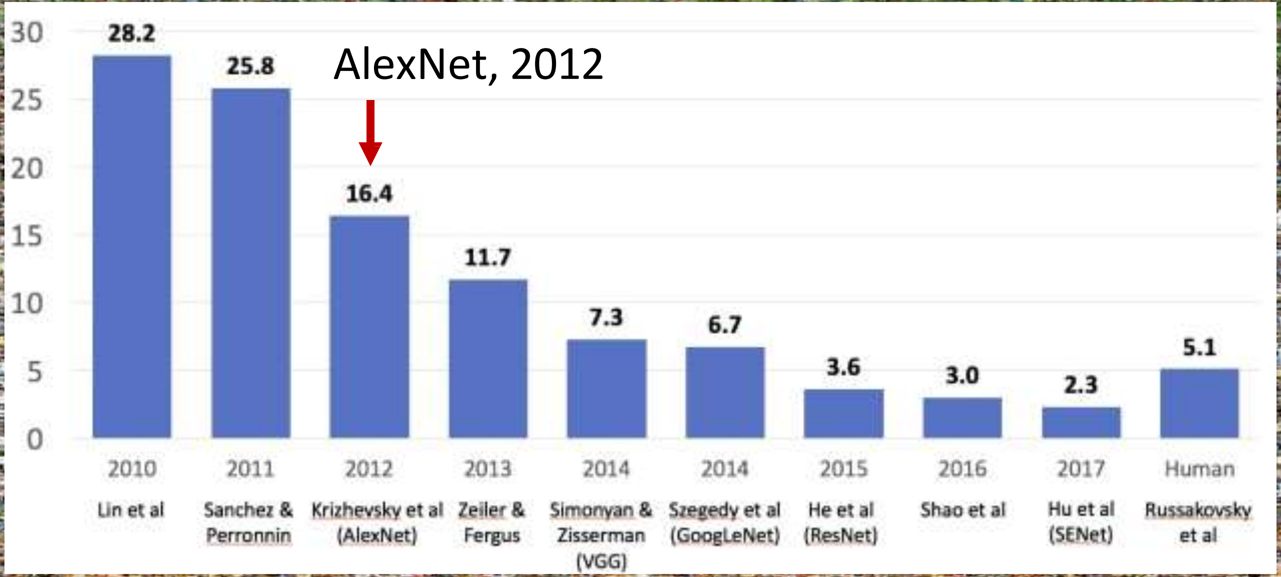
1958 Perceptron
1969 Minsky & Papert
1980 Neocognitron
1985 Backprop
1998 LeNet
2006 Deep Learning

IMAGENET Large Scale Visual Recognition Challenge





IMAGENET Large Scale Visual Recognition Challenge



1959 Hubel & Wiesel

1958 Perceptron

1963 Roberts

1969 Minsky & Papert

1970s David Marr

1979 Gen. Cylinders

1980 Neocognitron

1986 Canny

1985 Backprop

AI Winter

1997 Norm. Cuts

1998 LeNet

1999 SIFT

2001 V&J

2004, 2007 Caltech101; PASCAL

2006 Deep Learning

2009 ImageNet

2012 AlexNet

2012 to Present: Deep Learning is Everywhere

Image Classification

Image Retrieval

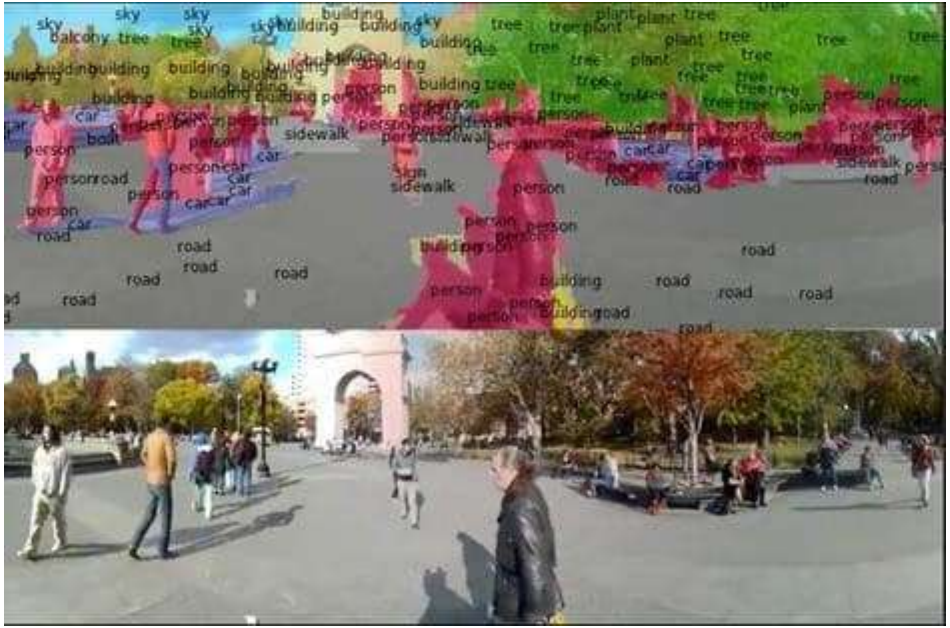
Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Object Detection

Ren, He, Girshick, and Sun, 2015

Image Segmentation

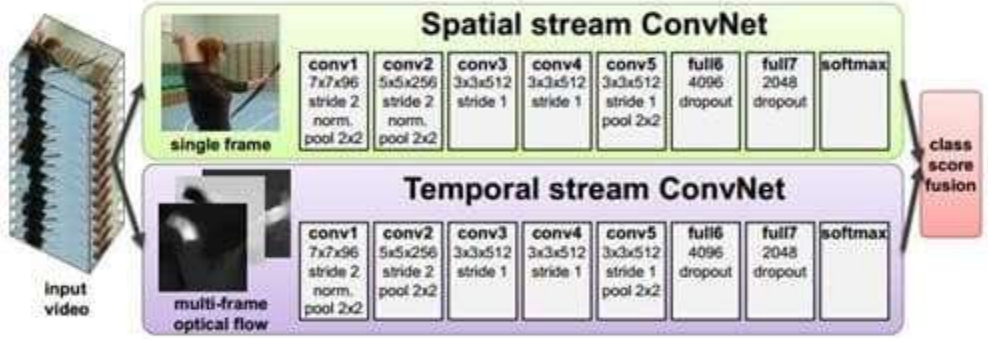


Fabaret et al, 2012

Slide inspiration: Justin Johnson

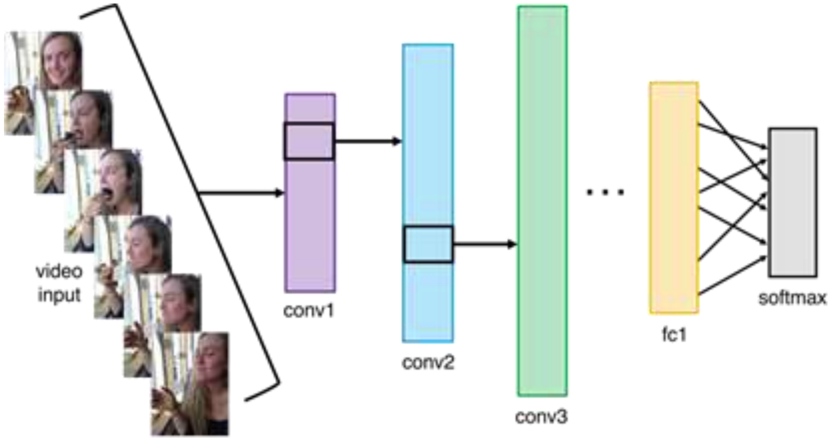
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition

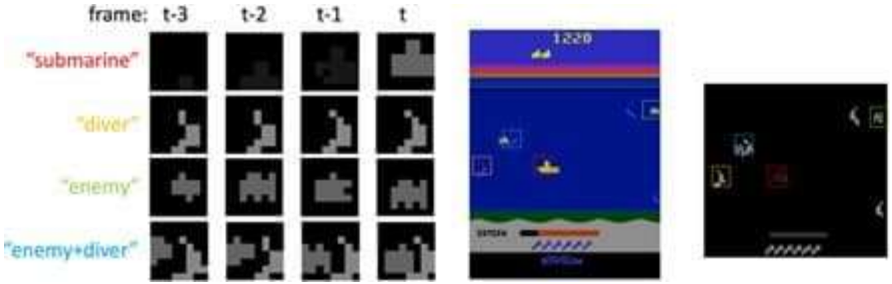


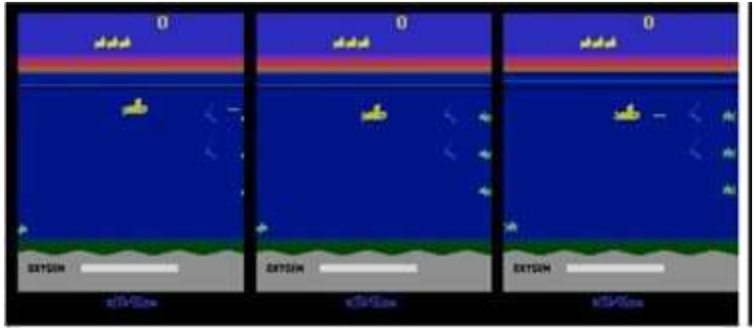
Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)

Playing Atari games (Guo et al, 2014)

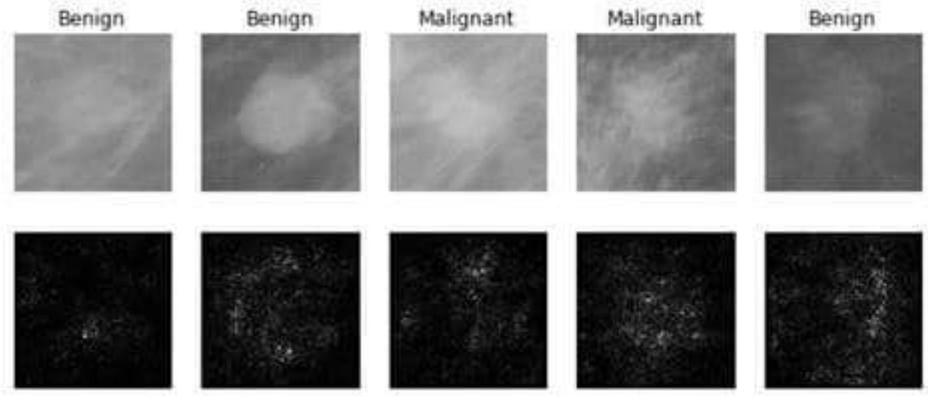




Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification

Dieleman et al, 2014

From left to right: public domain by NASA, usage permitted by ESA/Hubble, public domain by NASA, and public domain.

Whale recognition

[Kaggle Challenge]

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei, 2015

A white teddy bear sitting in the grass

A man in a baseball uniform throwing a ball

A woman is holding a cat in her hand

A man riding a wave on top of a surfboard

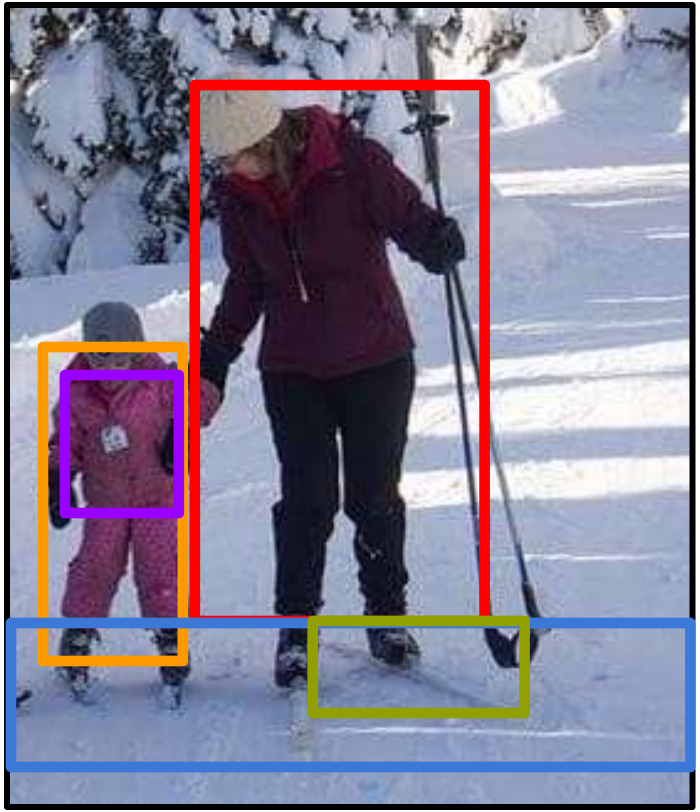
A cat sitting on a suitcase on the floor

A woman standing on a beach holding a surfboard

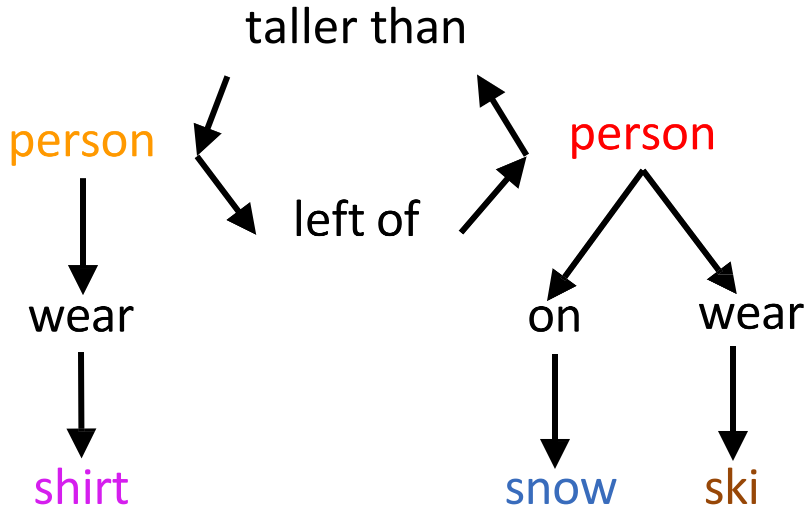
Captions generated by Justin Johnson using Neuraltalk2

Slide inspiration: Justin Johnson

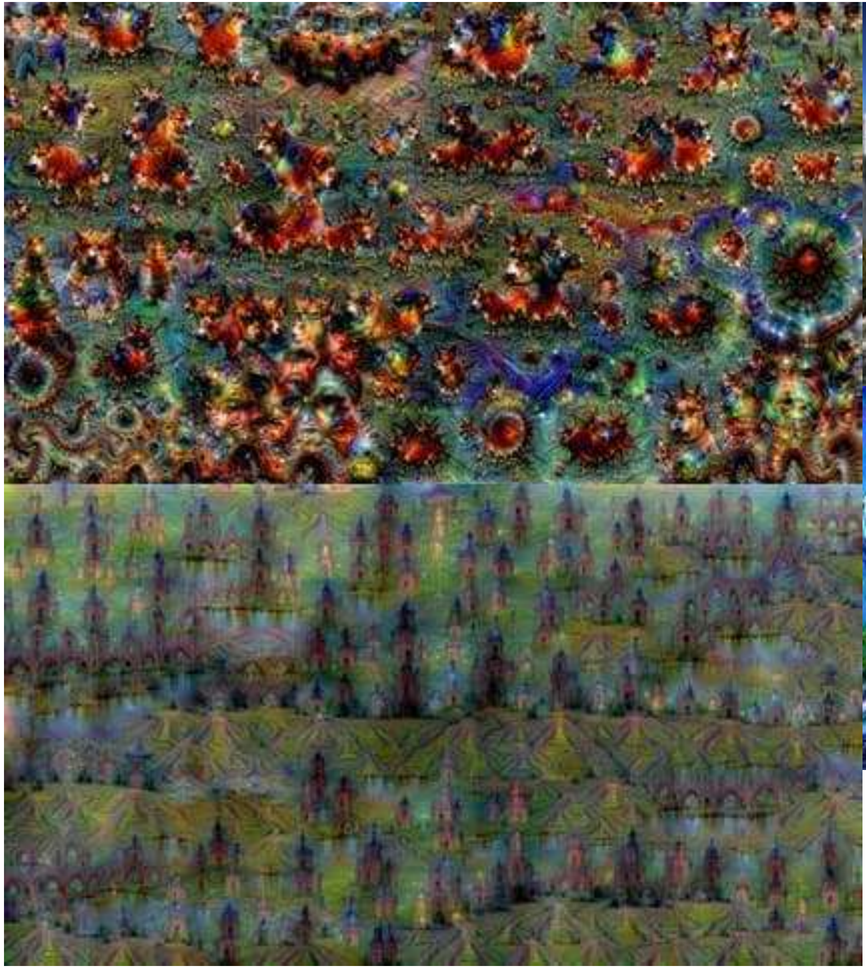
2012 to Present: Deep Learning is Everywhere



Results:
spatial, comparative, asymmetrical, verb, prepositional



Krishna*, Lu*, Bernstein, Fei-Fei, *ECCV* 2016



Mordvinsev et al, 2015
Gatys et al, 2016

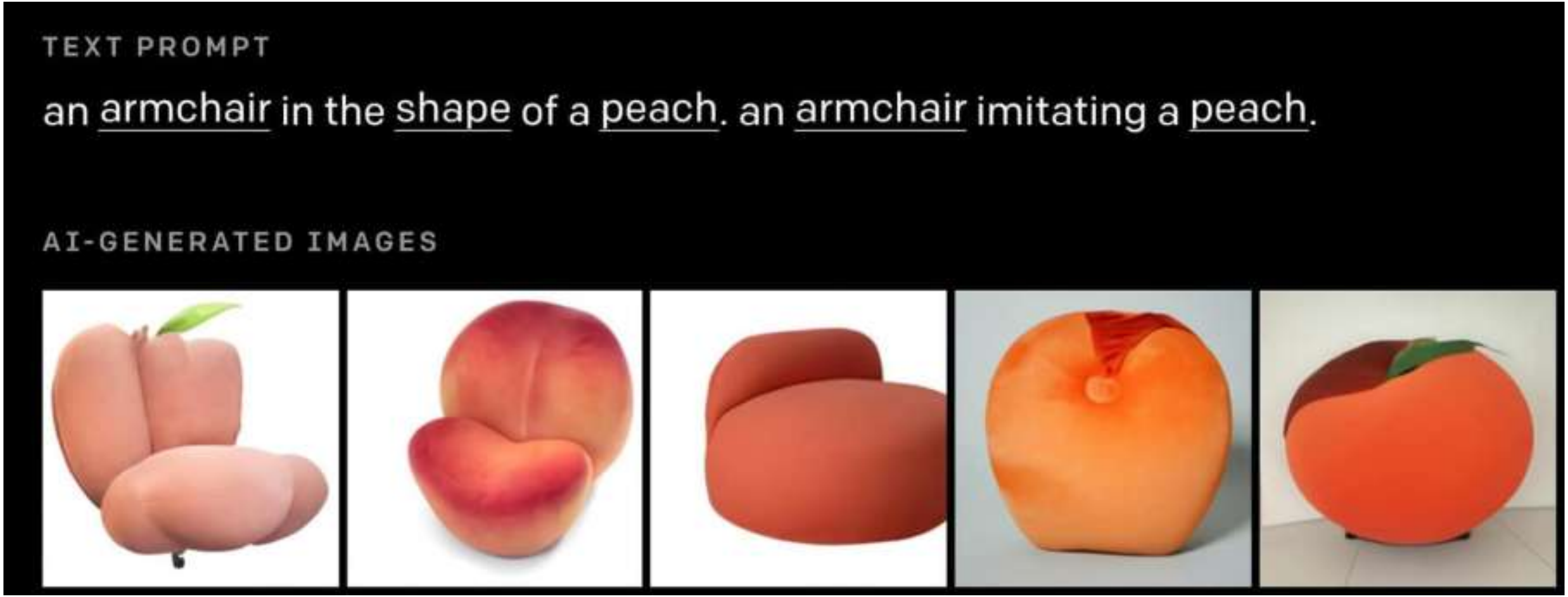
Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

Ramesh et al, “DALL·E: Creating Images from Text”, 2021. https://openai.com/blog/dall-e/

Slide inspiration: Justin Johnson

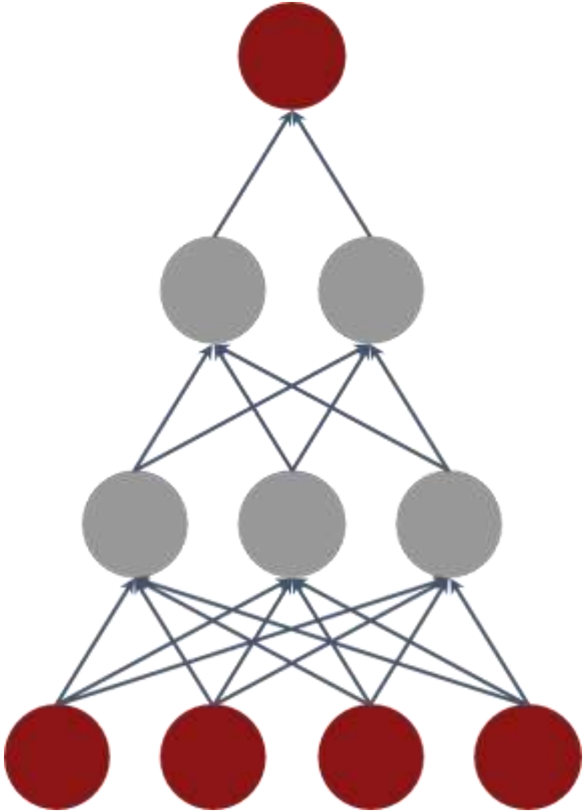
2012 to Present: Deep Learning is Everywhere



Slide inspiration: Justin Johnson

Ramesh et al, "DALL·E: Creating Images from Text", 2021. https://openai.com/blog/dall-e/

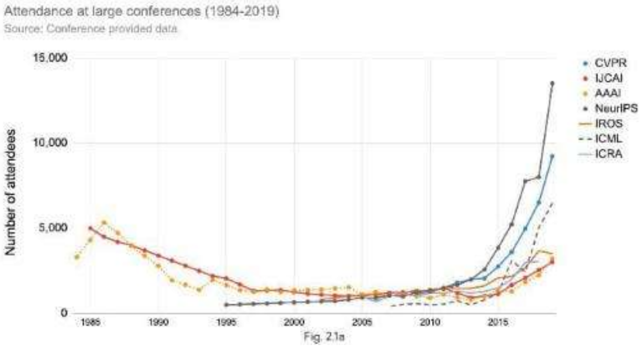
Computation



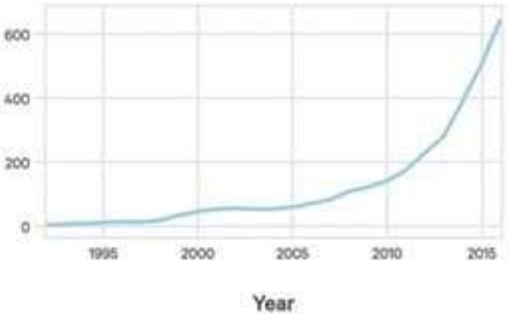
Algorithms

Data

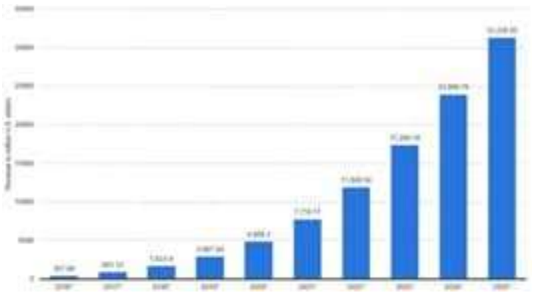
AI’s Explosive Growth & Impact



Number of attendance At AI conferences



Startups Developing AI Systems



Enterprise Application AI Revenue

Source: The Gradient

Source: Crunchbase, VentureSource, Sand Hill Econometrics

Source: Statista

Despite the successes, computer vision still has a long way to go

Computer Vision Can Cause Harm

Harmful Stereotypes

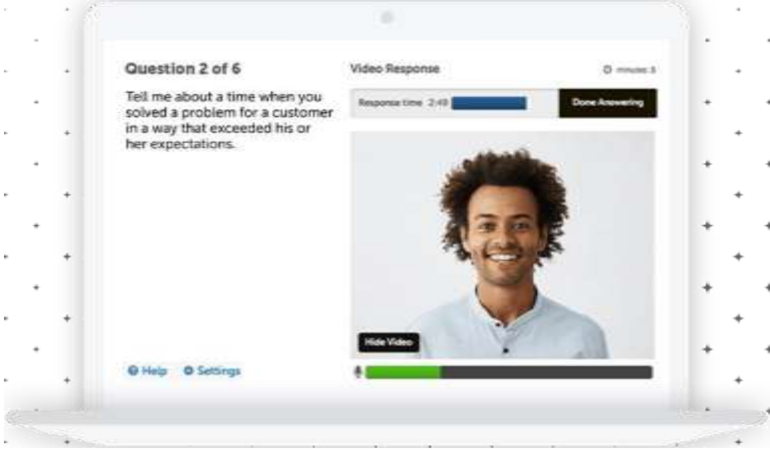
Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decid es-whether-you-deserve-job/
https://www.hirevue.com/platform/online-video-interviewing-software

Example Credit: Timnit Gebru

Computer Vision Can Save Lives

Slide inspiration: Justin Johnson

And there is a lot we don't know how to do

https://fedandfit.com/wp-conten t/uploads/2020/06/summer-activities-for-kids_optimized-scaled.jpeg

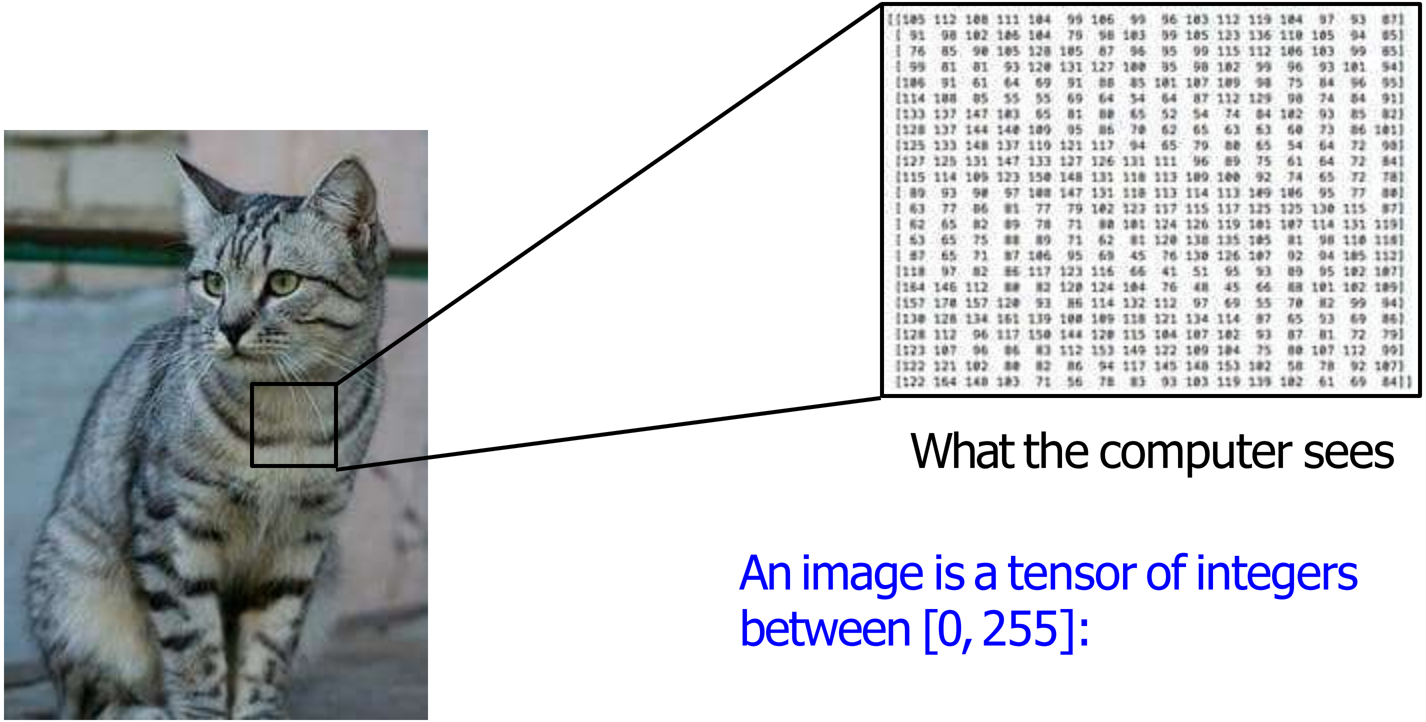
Slide inspiration: Andrej Karpathy

Image Classification: A core task in Computer Vision

(assume given a set of possible labels)
{dog, cat, truck, plane, ...}

$\longrightarrow$ cat

The Problem: Semantic Gap

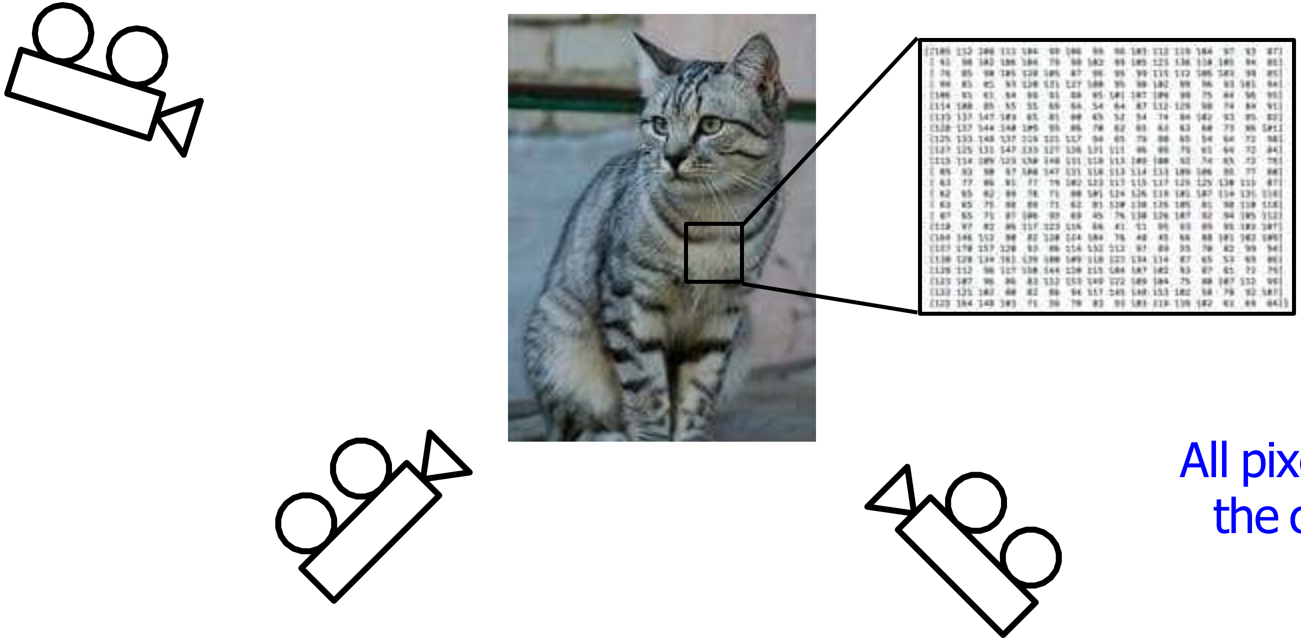


What the computer sees

An image is a tensor of integers between [0, 255]:

e.g. 800 x 600 x 3
(3 channels RGB)

Challenges: Viewpoint variation



All pixels change when the camera moves!

Challenges: Illumination

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Challenges: Background Clutter

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Challenges: Occlusion

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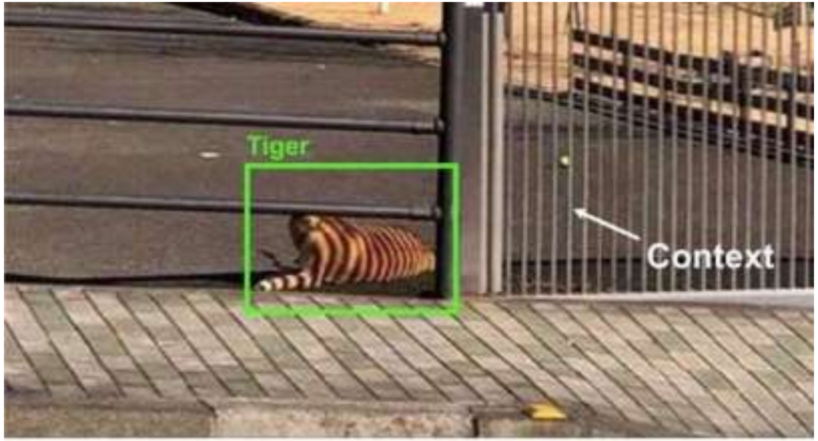
Challenges: Deformation

Challenges: Intraclass variation

Challenges: Context

Image source: https://www.linkedin.com/posts/ralph-aboujaoude-diaz-40838313_technology-artificialintelligence-computervision-activity-6912446088364875776-h-Iq?utm_source=linkedin_share&utm_medium=member_desktop_web

Challenges: Context



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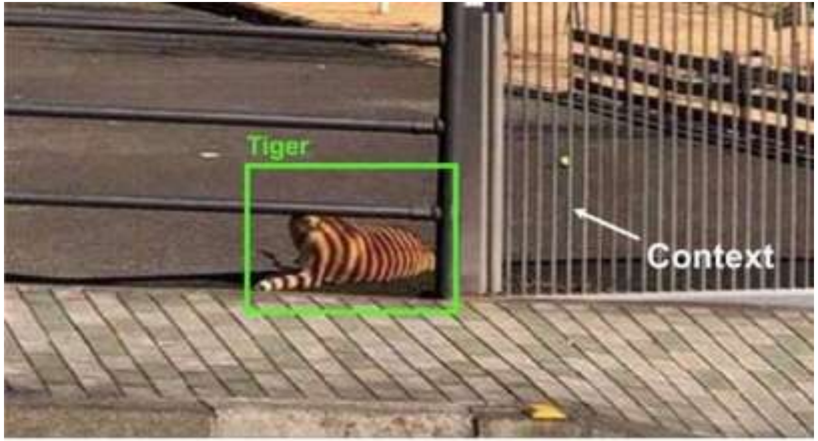

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Modern computer vision algorithms